

Aim:- To study constructors in Java & implementation of programs.

Theory:- A constructor is a special method in Java that is automatically called when an object is created.

Characteristics -

- Same name as class
- No return type
- Called automatically
- Used to initialize objects.

Types of Constructors -

- (1) Default Constructor
- (2) Parameterized Constructor
- (3) Copy Constructor
- (4) Constructor overloading.

(1) Default Constructor:-

A constructor with no parameters is called a default constructor.

- Provided by java automatically (if no constructor is written)
- or defined by programmer.
- It is called automatically.

output

2 5 8 11 14

Program 1: Demonstrate Default constructor with Number series.

```

class NumberSeries {
    NumberSeries() {
        int num = 2;
        for (int i = 0; i < 5; i++) {
            System.out.print(num + " ");
            num += 3;
        }
    }

    public static void main(String[] args) {
        new NumberSeries();
    }
}

```

2. Copy Constructor:-

Java doesn't have built-in copy constructor like C++, we can create it manually.

A constructor that copies values from another object.

```

class Demo (Demo obj) {
    num = obj.num;
}

```

```

Demo (int n) {
    num = n;
}

```

```

void display () {
    System.out.println("Value: " + num);
}

```

Teacher's Signature _____

Output
value : 10

Output

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

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```
public static void main (String[] args) {
    Demo d1 = new Demo (10);
    Demo d2 = new Demo (d1); // Copy
    d2.display();
}
}
```

(3) Parameterized Constructor

A constructor that accepts arguments (Parameters)

Program - Display Number Pattern with the help of Parameterized Constructor

```
Class NumberPattern {
    NumberPattern (int n) {
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= i; j++) {
                System.out.print (j + " ");
            }
            System.out.println();
        }
    }

    public static void main (String[] args) {
        new NumberPattern (5);
    }
}
```

Teacher's Signature _____

(A) Private Constructor

A constructor declared as private is used to restrict object creation.

used in - Singleton class

Utility classes.

Ex

```
class Demo {
```

```
    private Demo() {
```

```
        System.out.println("Private Constructor");
```

```
    }
```

```
    public static void main (String[] args) {
```

```
        // Demo d = new Demo(); not allowed.
```

```
    }
```

```
}
```

Constructor Overloading

When multiple constructor exist in the same class with different parameters.

Program Demonstrate Constructor overloading for Fibonacci & armstrong number.

```
class NumberOperations {
```

```
    NumberOperations(int n) {
```

```
        int a = 0; b = 1;
```

```
        System.out.print("Fibonacci Series: ");
```

```
        for (int i = 1; i <= n; i++) {
```

```
            System.out.print(a + " ");
```

```
            int c = a + b;
```

```
            a = b;
```

```
            b = c;
```

```
}
```

Teacher's Signature _____


```
    System.out.println();  
}
```

```
NumberOperation (int n, boolean isArmstrong) {  
    int original = n;
```

```
    int sum = 0;
```

```
    while (n > 0) {
```

```
        int digit = n % 10;
```

```
        sum += digit * digit * digit;
```

```
        n /= 10;
```

```
    }
```

```
    System.out.println("Armstrong check:");
```

```
    if (sum == original) {
```

```
        System.out.println(original + " is an  
        Armstrong number");
```

```
    } else {
```

```
        System.out.println(original + " is not an  
        Armstrong number");
```

```
    }
```

```
}
```

```
public static void main (String[] args) {
```

```
    new NumberOperation (7);
```

```
    new NumberOperation (153, true);
```

```
}
```

```
}
```